Smart Grid Energy Management System

Technical Design Documentation

SYSTEM TYPE	APPLICATION DOMAIN	CONTROL STRATEGY
Embedded Control System	Industrial Power Management	Tiered Threshold Load Shedding

1. Project Overview

This project simulates a Smart Grid Energy Management System (EMS) designed to monitor total power consumption across three primary consumption nodes: **House**, **Factory**, and **Industry**. The system implements a tiered threshold strategy to maintain grid stability through continuous monitoring and automated load management.

When power usage exceeds predefined safety limits, the system issues operational warnings and systematically sheds lower-priority loads to prevent total power from exceeding the absolute substation limit. This approach ensures critical infrastructure remains operational during peak demand events while protecting grid infrastructure from overload conditions.

> **Primary Objective:** Maintain grid stability by keeping total power consumption below the substation limit of 2500 W through automated, priority-based load management.

2. System Architecture

The grid topology consists of three consumption nodes connected to a central substation. Each node contains multiple loads with varying criticality levels. The architecture is designed with clear hierarchical relationships between the substation, nodes, and individual loads.

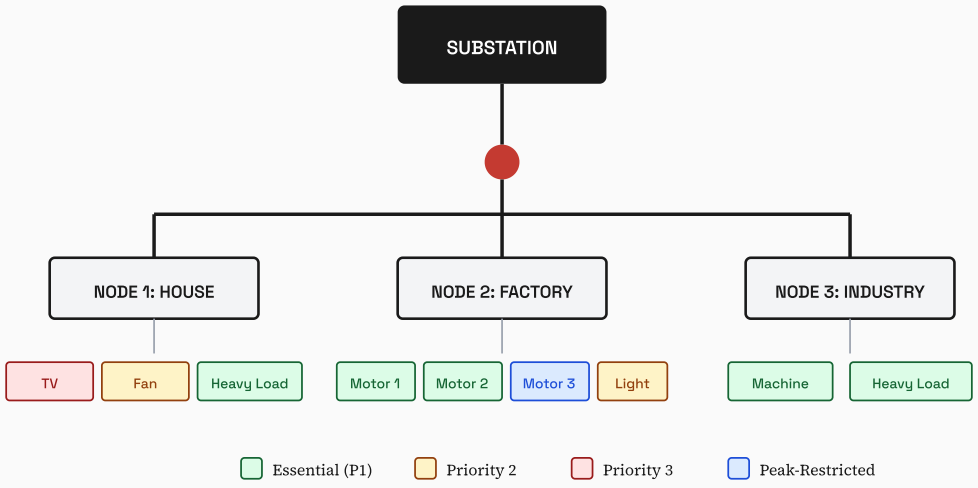

Figure 1: System topology showing node hierarchy and load classifications

2.1 Node Composition

NODE	LOAD NAME	CLASSIFICATION	BEHAVIOR
House	TV	P3	First-tier shedding
	Fan	P2	Second-tier shedding
	Heavy Load	P1	Never shed
Factory	Motor 1	P1	Essential
	Motor 2	P1	Essential
	Motor 3	P1	Peak-hour restricted
	Light	P2	Second-tier shedding
Industry	Machine	P1	Essential
	Heavy Load	P1	Essential

3. Priority Classifications

To maintain grid integrity during high-demand events, loads are classified into shed-able priority tiers. This classification determines the sequence in which non-essential loads are disconnected during overload conditions.

> **Note:** All loads not explicitly assigned to Priority 2 or Priority 3 are classified as Priority 1 (Essential) and are never automatically shed by the EMS, regardless of grid state.

3.1 Priority Tier Definitions

PRIORITY LEVEL	SHED SEQUENCE	LOADS	RATIONALE
P3	First (immediate)	House: TV	Non-essential comfort load; lowest impact on operations
P2	Second (cascading)	House: Fan Factory: Light	Secondary comfort loads; minimal operational disruption
P1	Never shed	All other loads	Critical infrastructure; essential operations

4. System Parameters and Thresholds

The EMS monitors the aggregate power consumption (`TotPwr`) against defined hardware limits and calculated safety thresholds. These thresholds create a graduated response system that prevents sudden grid failure.

4.1 Threshold Definitions

PARAMETER	SYMBOL	VALUE	CALCULATION
Substation Limit	`SubLim`	2500 W	Absolute hardware maximum
Warning Threshold	`SafeLim_1`	2125 W	0.85 × SubLim
Critical Threshold	`SafeLim_2`	2375 W	0.95 × SubLim

4.2 Threshold Visualization

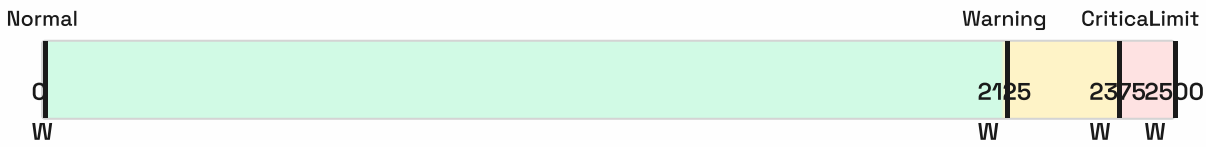

Figure 2: Power consumption zones and threshold boundaries

> **Safety Margin:** The 125 W gap between SafeLim_2 (2375 W) and SubLim (2500 W) provides a buffer zone for system response latency and measurement uncertainty.

5. Core Control Logic

The EMS Function Block executes logical routines continuously during every scan cycle. The control logic comprises three primary modules: power calculation, time-of-use management, and load shedding execution.

5.1 Total Power Calculation

The system calculates real-time power draw by summing the active wattage of all nodes currently in operation:

$$\mathrm{TotPwr} = \mathrm{House}_{\mathrm{loads}} + \mathrm{Factory}_{\mathrm{loads}} + \mathrm{Industry}_{\mathrm{loads}}$$

Each load contributes its rated wattage to the total only when in the "ON" state. The calculation is performed continuously to ensure real-time accuracy.

5.2 Time-of-Use Peak Management

The system monitors the current time of day to restrict high-draw equipment during expensive or high-demand periods. This proactive measure reduces grid strain during predictable peak windows.

> **Peak Hour Rule:** When operating within a designated Peak Hour window, Factory Motor 3 is disabled and cannot be activated, regardless of total grid capacity. This restriction applies before any threshold-based load shedding.

5.3 Load Shedding Logic Flow

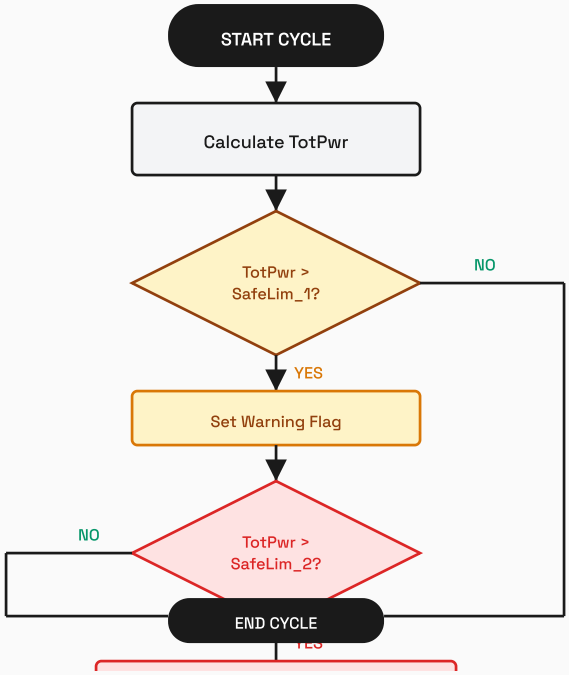

Figure 3: Control logic flow diagram

6. Load Shedding Sequence

The EMS evaluates TotPwr against safety limits in a cascading sequence to ensure the grid does not exceed 2500 W. The shedding logic operates on a priority basis, removing lowest-priority loads first.

6.1 Level 1: Warning State

CONDITION	ACTION	SYSTEM RESPONSE
`TotPwr > SafeLim_1` (> 2125 W)	Set Warning Flag	● HMI alarm triggered No loads disconnected System enters alert state

6.2 Level 2: P3 Shedding

CONDITION	ACTION	LOADS AFFECTED
`TotPwr > SafeLim_2` (> 2375 W)	Shed Priority 3 loads	● House TV → OFF Immediate disconnection

6.3 Level 3: P2 Shedding

CONDITION	ACTION	LOADS AFFECTED
`TotPwr > SafeLim_2` (after P3 shed)	Shed Priority 2 loads	● House Fan → OFF Factory Light → OFF System stabilizes

7. Sequence of Operations Example

The following scenario demonstrates the EMS behavior under increasing load conditions, illustrating the tiered response mechanism.

1 **Initial State:** Operator activates House Heavy Load, Factory Motors 1 & 2, and Industry Machine.
`TotPwr = 2000 W`
● System Status: **Normal State**

2 **Warning Trigger:** Operator activates Factory Motor 3.
`TotPwr = 2200 W` (exceeds SafeLim_1)
● System Status: **Warning State** — HMI alarm active

3 **Critical Threshold:** Operator activates Industry Heavy Load.
`TotPwr = 2450 W` (exceeds SafeLim_2)
● System Status: **Critical State**

4 **Automatic P3 Shedding:** System forces House TV OFF.
`TotPwr = 2400 W` (still above SafeLim_2)
● System Status: **Shedding in Progress**

5 **Automatic P2 Shedding:** System forces House Fan and Factory Light OFF.
`TotPwr = 2300 W` (below SafeLim_2)
● System Status: **Stabilized — Warning State**

7.1 State Transition Summary

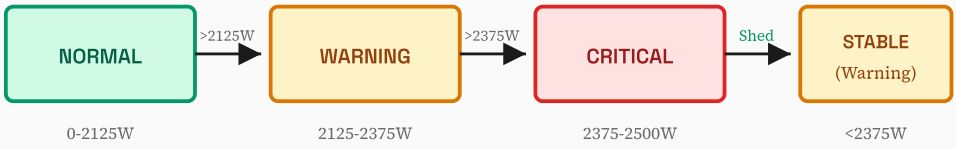

Figure 4: System state transitions based on power consumption levels

8. Implementation Notes

8.1 Scan Cycle Considerations

The EMS logic executes within the controller's scan cycle. For optimal performance:

- Power calculations should be performed at the beginning of each cycle
- Load shedding commands should be issued before cycle completion
- Adequate delay should be incorporated between shedding stages to allow power stabilization

8.2 Safety Constraints

> **Critical:** The system must never allow TotPwr to exceed SubLim (2500 W). The tiered threshold approach ensures multiple intervention points before reaching the absolute limit.

- Essential loads (P1) are protected from automatic shedding under all conditions
- Manual override capability should be restricted to authorized personnel only
- System must log all shedding events with timestamps for audit purposes